

Pinlight AR Near-Eye Display with Large FOV and Expanded Eyebbox through Eye Tracking

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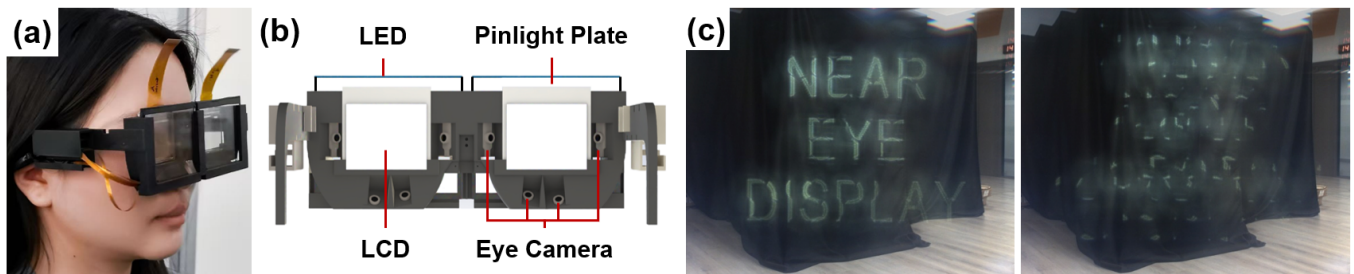


Figure 1: Overview of wide-field-of-view pinlight display with dynamic eyebbox steering. (a) Implemented wearable prototype. (b) Designed 3D model of the proposed prototype. (c) Comparison of perceived images. The left shows results driven by our proposed Learning-based 3D Pupil Localization, maintaining image clarity. The right shows results driven by traditional geometric triangulation, where tracking errors lead to unclear image.

Abstract

We have developed a novel design of a compact pinlight near-eye display for augmented reality (AR) featuring a wide field of view (FOV) and an expanded eyebbox through dynamic eye tracking. In our design, a camera rig mounted on the pinlight AR NED captures multi-view observations of the viewer's pupil, and the captured

pupil images are utilized to predict the 3D pupil location through a trained neural network. The 3D pupil location is then used to calculate a display pattern that spatio-temporally modulates the light entering the viewer's eyes. A prototype following the design is developed, with preliminary testing conducted, showing the effectiveness of our proposed method.

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SIGGRAPH Emerging Technologies '26, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-2549-4/26/07

<https://doi.org/10.1145/3799826.3812432>

CCS Concepts

• Hardware → Displays and imagers; • Computing methodologies → Mixed / augmented reality.

ACM Reference Format:

Zheye Yu, Shenghao Zhao, Jing Zhang, Xinxing Xia, Andrei State, Tat-Jen Cham, Frank Guan, and Henry Fuchs. 2026. Pinlight AR Near-Eye Display with Large FOV and Expanded Eyebbox through Eye Tracking. In *Special Interest Group on Computer Graphics and Interactive Techniques*

Conference Emerging Technologies (SIGGRAPH Emerging Technologies '26), July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3799826.3812432>

1 Introduction

Optical-see-through (OST) near-eye displays (NEDs) are pivotal for augmented reality (AR), but remain constrained by trade-offs between field-of-view (FOV), eyebox size, resolution and form factor [Itoh et al. 2021]. Pinlight AR NEDs (PNEDs) show promise for optical-see-through AR NEDs due to their large FOV, compact form factor, and low manufacturing cost [Maimone et al. 2014; Xia et al. 2025]. However, PNEDs suffer from a minuscule eyebox, a shortcoming that makes them unusable as wide-FOV NEDs in practical scenarios. In this work, we present a novel design that expands the eyebox for PNEDs. We utilize a quad-camera rig to extract multi-view 2D pupil features of each viewer's eye, which are then processed by a trained neural network to accurately predict the 3D pupil location. The predicted 3D pupil location will then be used to dynamically update the modulating patterns in PNEDs to increase display quality for the viewer. This data-driven approach towards eyebox expansion overcomes limitations of geometric solvers, ensuring a stable viewing experience and allowing viewers to fully enjoy the PNED's inherently large FOV (see Fig. 1).

2 Our Method

2.1 Design

Our design integrates three primary components (Fig. 2): a pinlight plane, a transparent modulation plane, and an eye tracking camera rig. The pinlight plane is a transparent panel featuring a dense array of light guide dots, with edge-mounted LEDs generating a field of point light sources. These emitted light beams traverse the transparent modulation plane (in our case, a transparent LCD), where individual modulation elements regulate the intensity of specific light rays before they enter the viewer's eye to form an image on the retina. To enable active eyebox steering, an eye-tracking camera rig comprising four cameras is mounted and oriented toward the viewer's eye. The captured images of the viewer's pupil are then used to predict the pupil's 3D position through a trained neural network.

Figure 2 illustrates how the modulation pattern on the transparent LCD changes when the pupil location changes. At time t_1 , the eye-tracking camera rig captures four sets of eye images and the neural network predicts the corresponding 3D pupil position, which is then used to update the LCD modulation pattern so that the eye perceives an image "E" at the focus plane. At time t_2 , suppose the eyeball has rotated to a new location, and so a new pattern must be computed and displayed in the LCD, enabling the eye to continue seeing the same "E" at the focus plane.

2.2 Learning-based 3D Pupil Localization

A problem arises when the AR NED is moved closer to the viewer's eye (e.g., with eye relief at 15 mm to increase the FOV as well as reduce the form factor). Here, the tracking cameras may not simultaneously capture complete images of the pupils at all times, due to self-occlusion when the eyeball rotates away from a frontal direction. To robustly handle such instances when the pupil is

partially or fully occluded in a subset of the cameras, we developed a learning-based method for localizing the 3D pupil position with higher accuracy than can be achieved with classical approaches.

In our method (Fig. 3), two industrial cameras are used to calculate ground-truth 3D pupil locations for network training. We mount a camera rig comprising four cameras in front of the viewer's eye, and calibrated the two industrial cameras and four mounted cameras to each other. For each of the four cameras, we extract the 2D pupil centroid along with its detection confidence score from EllSeg ([Kothari et al. 2021]). We aggregate the information by flattening the inputs from all four cameras into a 12-dimensional concatenated feature vector. This vector is fed into a lightweight end-to-end regression network consisting of a Multi-Layer Perceptron (MLP) with three fully-connected layers. By supervised training with the 3D ground-truth locations, the network implicitly learns to weigh the contributions of each camera to intelligently handle images with partially visible pupils.

3 Prototype, Validation and Demonstration

A prototype of our PNED with a dynamically expanded eyebox is shown in Fig. 1 (a) and (b). The pinlight plane is fabricated from a 2 mm thin acrylic sheet with laser-drilled tiny divots, and is illuminated by LEDs on its top edge. A 5 mm thin transparent LCD panel located 14 mm in front of the pinlight plane. An eyeglasses-like frame has been 3D printed and four cameras are mounted for each eye. For a typical eye-to-LCD distance of 15 mm, the resulting horizontal FOV is 93° while the vertical FOV is 75°.

We validated our PNED performance on cases when the viewer's pupil location cannot be calculated with geometry-based method. For example, Fig. 1 (c) shows the obtained results using our method in comparison with the geometry-based method when the viewer's pupil is set to be 15 mm from the LCD display and the viewer's pupil is only detectable by two out of four tracking cameras. We validated our PNED performance on cases when the viewer's pupil location cannot be calculated with geometry-based method. For example, Fig. 1 (c) shows the obtained results using our method in comparison with the geometry-based method when the viewer's pupil is set to be 15 mm from the LCD display and the viewer's pupil is only detectable by two out of four tracking cameras.

During the demonstration, the participant is seated and wears the proposed PNED. A large television is positioned about 1 m in front of the participant and is viewed through the PNED. A bright dot is displayed on the television, and the participant is instructed to track the dot with their gaze. Meanwhile, a virtual image is presented by the PNED. As the participant's eye moves to different positions, the virtual image is perceived clearly by the participant.

4 Future Work

In future work, we plan to scale up the training dataset with a large number of participants and diverse eye poses and appearances. This will improve the generalization of the learning-based 3D pupil localization model across users. We also aim to explore the data capturing with different cameras, e.g., meta-lens based cameras, to investigate the possibility of eye-tracking with fewer cameras.

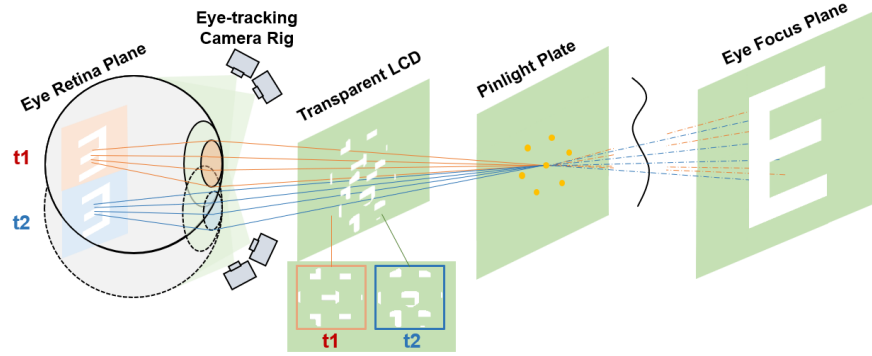


Figure 2: The principle of our proposed pinlight AR NED with pupil localization.

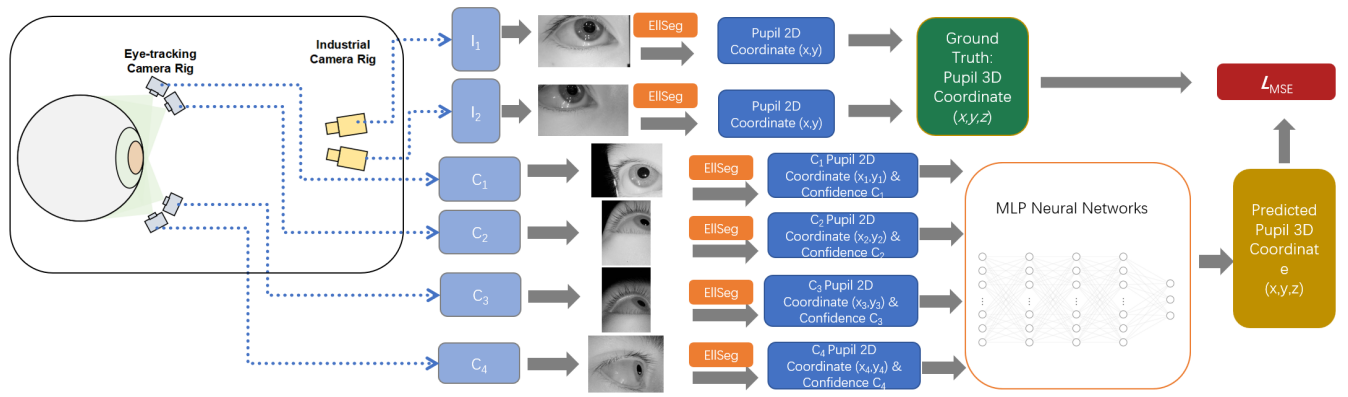


Figure 3: Overview of the pipeline of learning-based pupil 3D localization

Acknowledgments

This project is supported by the National Science Foundation of China (62575167), the Science and Technology Commission of Shanghai Municipality (24511106502), the MOE Academic Research Fund Tier 1 of Singapore, and partially supported by the MOE Translational and Innovation Fund (TIF) (Award Number: MOE2023_TIF-0006).

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